

GLA UNIVERSITY- MATHURA

DSA ASSIGNMENT-1

Name- Aditya Agarwal

Section – FB

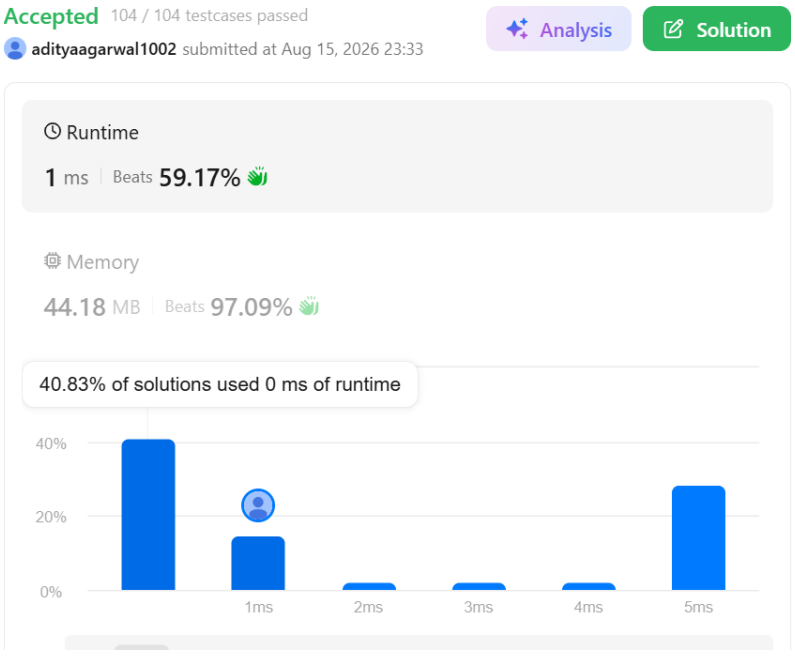
University Roll Number- 12515000088

Class Roll No.-03

Q1464. MAXIMUM PRODUCTS OF TWO ELEMENTS IN AN ARRAY

Java Auto

```java
class Solution {
    public int maxProduct(int[] nums) {
        int max1=0;
        int max2=0;
        for(int i=0;i <nums.length;i++){
            if(nums[i]>max1){
                max2=max1;
                max1=nums[i];
            }
            else if(nums[i]>max2){
                max2=nums[i];
            }
        }
        return(max1-1)*(max2-1);
    }
}
```

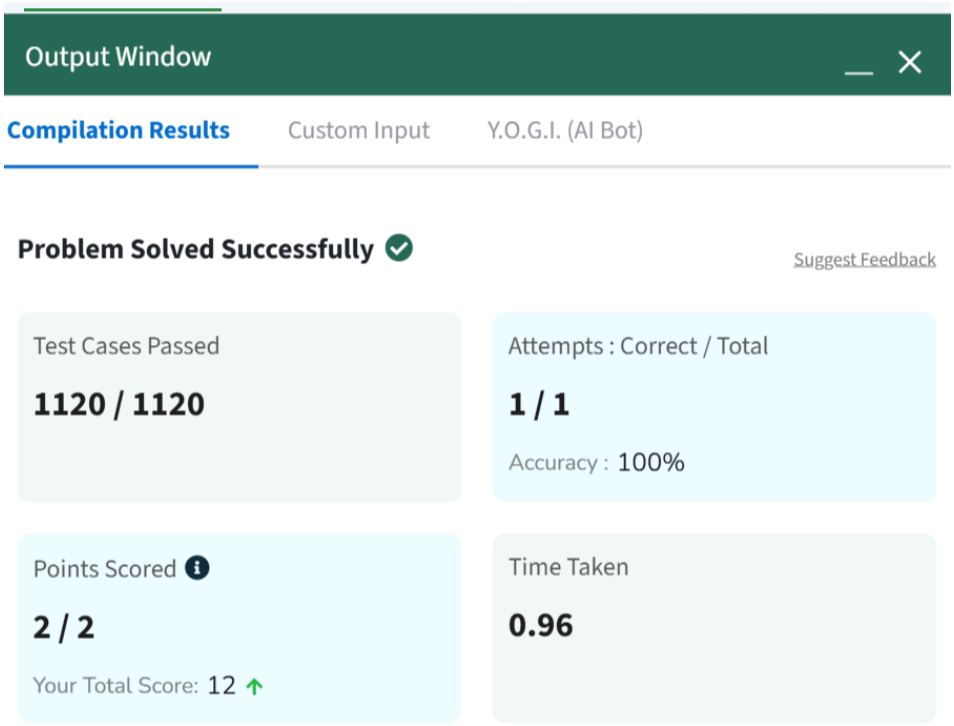


Time Complexity : O(n)

Space Complexity : O(1)

Q . SECOND LARGEST

```java
class Solution {
    public int getSecondLargest(int[] arr) {
        // code here
        Arrays.sort(arr);
        for(int i=arr.length-2;i>=0;i--){
            if(arr[i] != arr[arr.length-1]){
                return arr[i];
            }
        }
        return -1;
    }
}
```



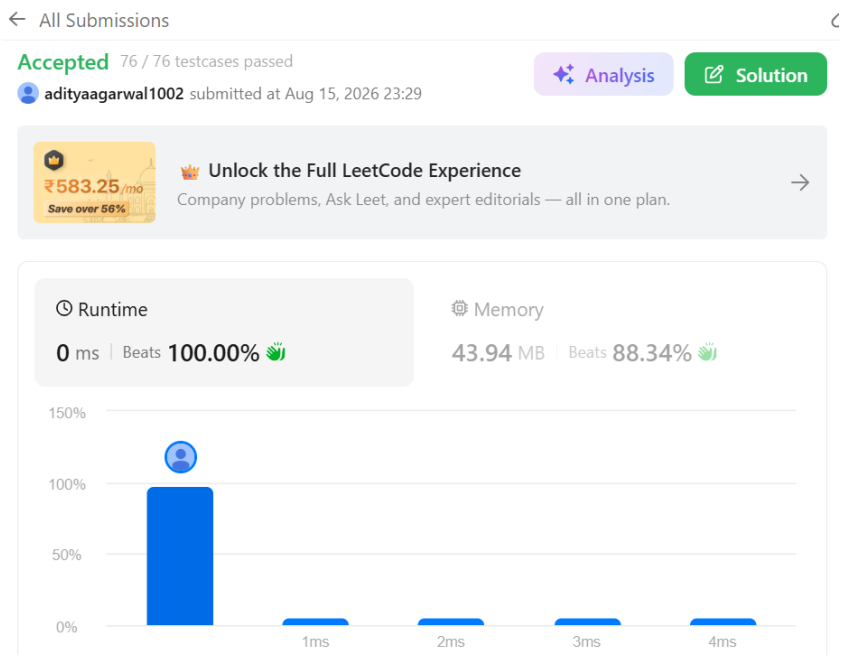
Time Complexity : O(n log n)

Space Complexity : O(log n)

Q 162. FIND PEAK ELEMENT

Java Auto

```java
class Solution {
    public int findPeakElement(int[] nums) {
        int left =0;
        int right=nums.length-1;
        while(left<right){
            int mid=(left+right)/2;
            if(nums[mid]>nums[mid+1]){
                right=mid;
            } else{
                left=mid+1;
            }
        }
        return left;
    }
}
```



Time Complexity : O(log n)

Space Complexity : O(1)

Q 33. SEARCH IN ROTATED SORTED ARRAY

Java Auto

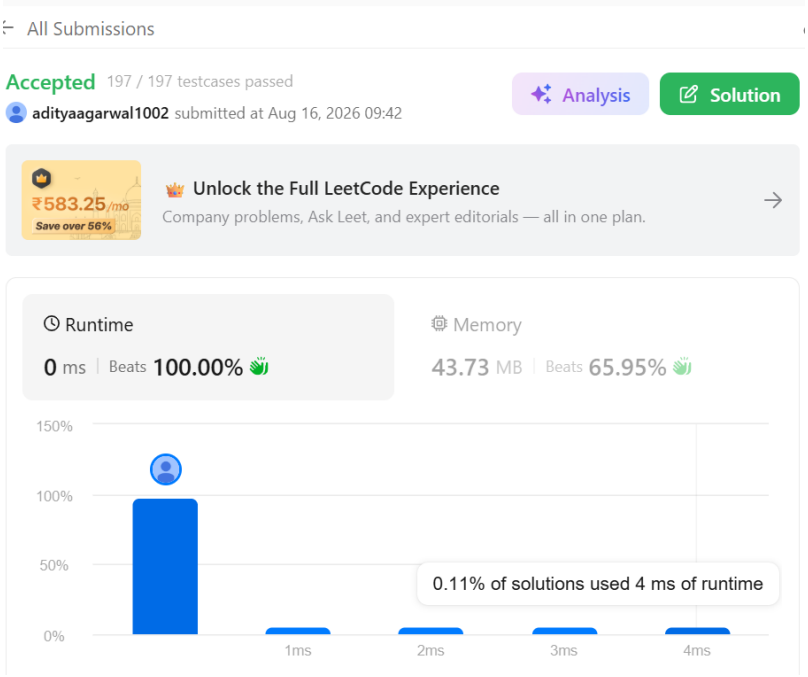
```java
class Solution {
    public int search(int[] nums, int target) {

        int left = 0;
        int right = nums.length - 1;

        while (left <= right) {
            int mid = (left + right) / 2;

            if (nums[mid] == target)
                return mid;

            if (nums[left] <= nums[mid]) {
                if (target >= nums[left] && target < nums[mid])
                    right = mid - 1;
                else
                    left = mid + 1;
            } else {
                if (target > nums[mid] && target <= nums[right])
                    left = mid + 1;
                else
                    right = mid - 1;
            }
        }
```



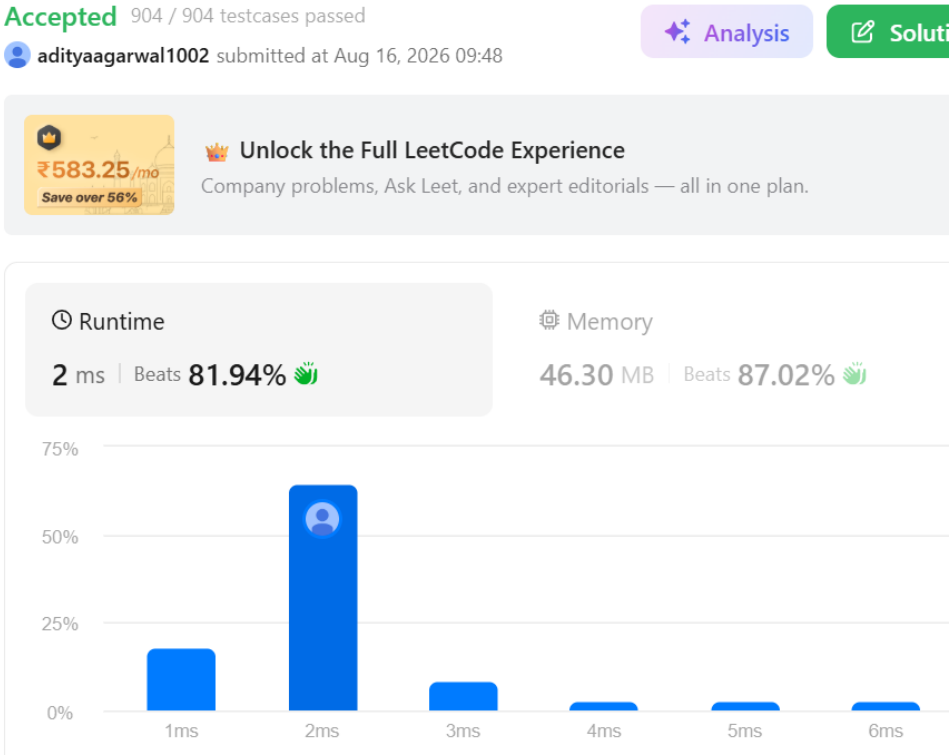
Time Complexity : O(log n)

Space Complexity : O(1)

Q.3477 FRUITS INTO BASKETS II

Java Auto

```java
class Solution {
    public int numOfUnplacedFruits(int[] fruits, int[] baskets) {
        int count = 0;
        for (int i = 0; i < fruits.length; i++) {
            for (int j = 0; j < baskets.length; j++) {
                if (baskets[j] >= fruits[i]) {
                    baskets[j] = 0;
                    break;
                }
                if (j == baskets.length - 1)
                    count++;
            }
        }
        return count;
    }
}
```



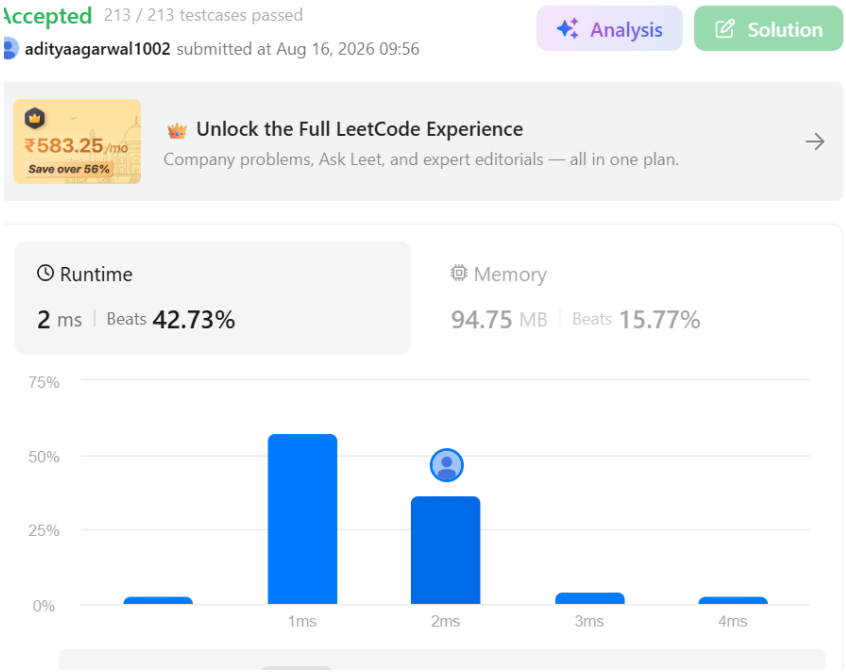
Time Complexity : $O(n^2)$

Space Complexity : O(1)

Q.121. BEST TIME TO BUY AND SELL STOCK

Java Auto

```java
class Solution {
    public int maxProfit(int[] prices) {
        int min = prices[0];
        int profit = 0;
        for (int i = 1; i < prices.length; i++) {
            if (prices[i] < min) {
                min = prices[i];
            }
            if (prices[i] - min > profit) {
                profit = prices[i] - min;
            }
        }
        return profit;
    }
}
```

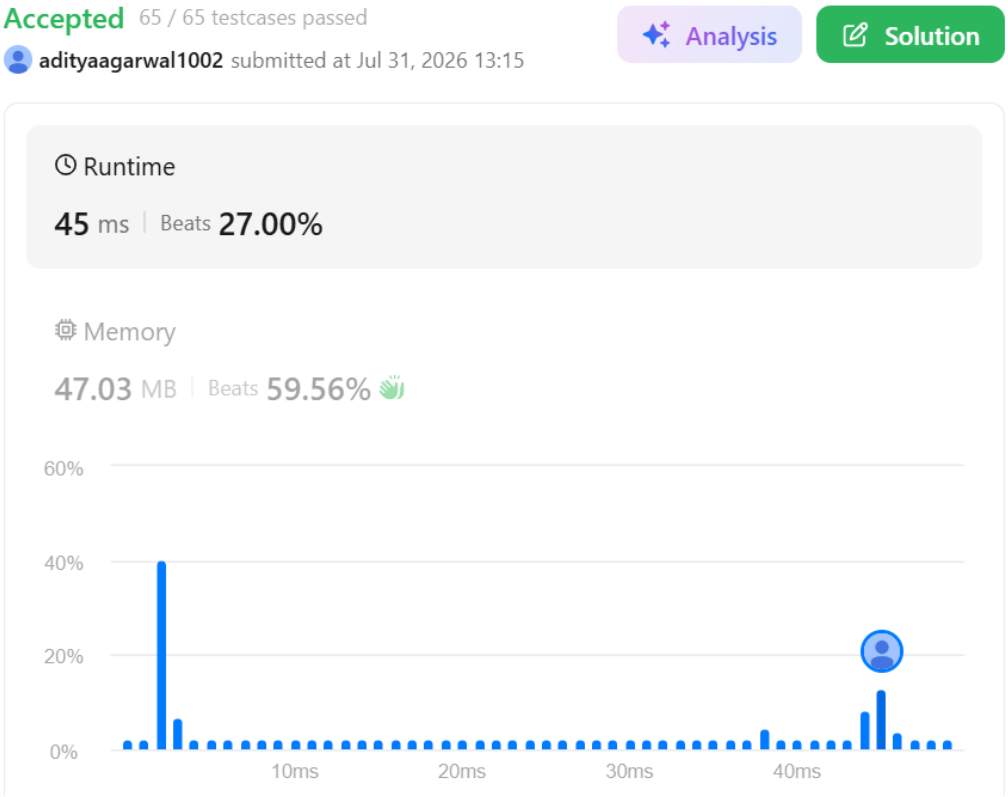


Time Complexity : O(n)

Space Complexity : O(1)

Q.1 TWO SUM

```java
class Solution {
    public int[] twoSum(int[] num, int target) {
        for(int i=0; i<num.length-1; i++){
            for(int j=i+1; j<num.length; j++)
            if(num[i] + num[j] == target){
                return new int[]{i,j};
            }
        }
        return new int[]{-1,-1};
    }
}
```



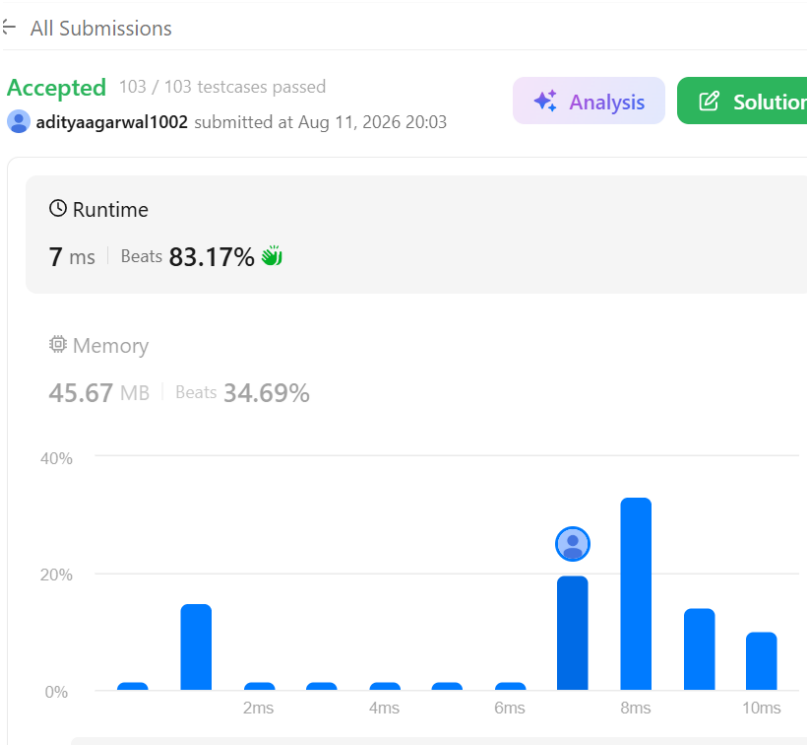
Time Complexity : $O(n^2)$

Space Complexity : $O(1)$

Q.1365 HOW MANY NUMBERS ARE SMALLER THAN A CURRENT NUMBER

```java
class Solution {
    public int[] smallerNumbersThanCurrent(int[] nums) {
        int[] ans =new int[nums.length];
        for(int i=0;i< nums.length;i++){
            int count=0;
            for(int j=0;j< nums.length;j++){
                if(nums[j]<nums[i]){
                    count++;
                }
            }
            ans[i]=count;;
        }
        return ans;

    }
}
```



Time Complexity : $O(n^2)$

Space Complexity : $O(n)$

Q. 628 MAXIMUM PRODUST OF THREE NUMBERS

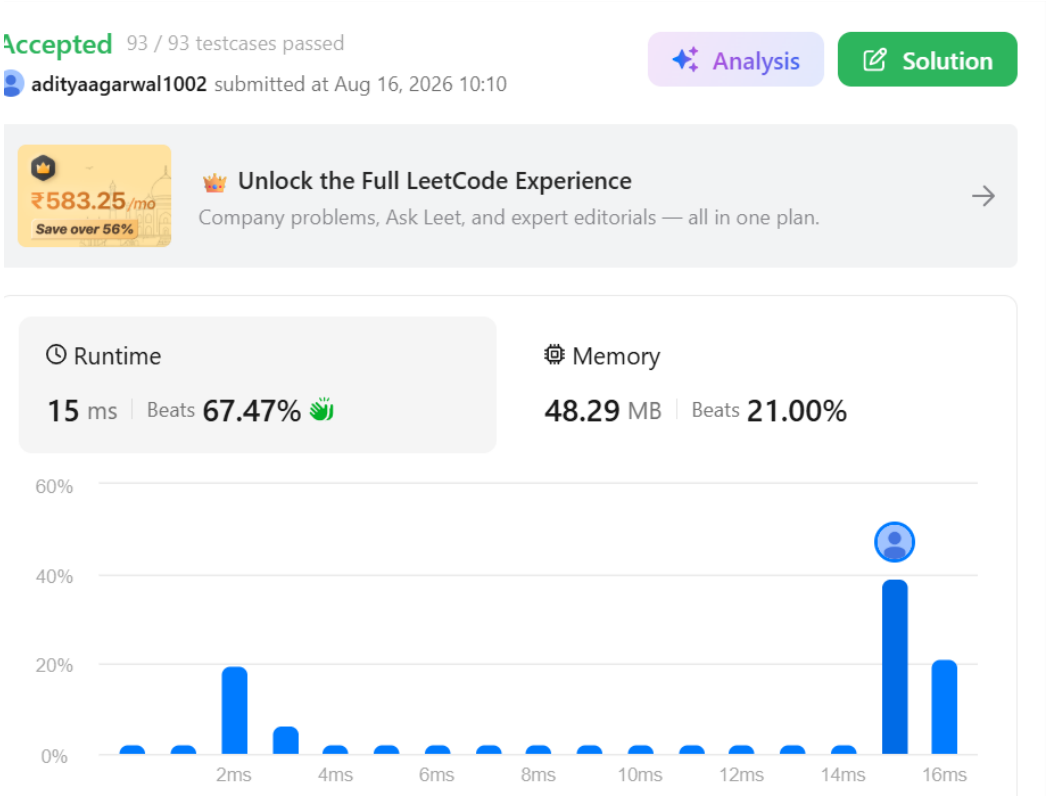
Java Auto

```java
import java.util.Arrays;
class Solution {
    public int maximumProduct(int[] nums) {
        Arrays.sort(nums);
        int n = nums.length;

        int a = nums[n-1] * nums[n-2] * nums[n-3];
        int b = nums[0] * nums[1] * nums[n-1];

        return Math.max(a, b);
```

Saved



Time Complexity : O(n log n)

Space Complexity : O(1)

Q .26 REMOVE DUPLICATES FROM SORTED ARRAY

ava Auto

```java
class Solution {
    public int removeDuplicates(int[] arr) {
        int j=0;
        for(int i=1;i<arr.length;i++){
            if(arr[i]!=arr[j]){
                j++;
                arr[j]=arr[i];
            }
        }
        return j+1;
    }
}
```

Accepted 362 / 362 testcases passed

adityaagarwal1002 submitted at Aug 13, 2026 13:56

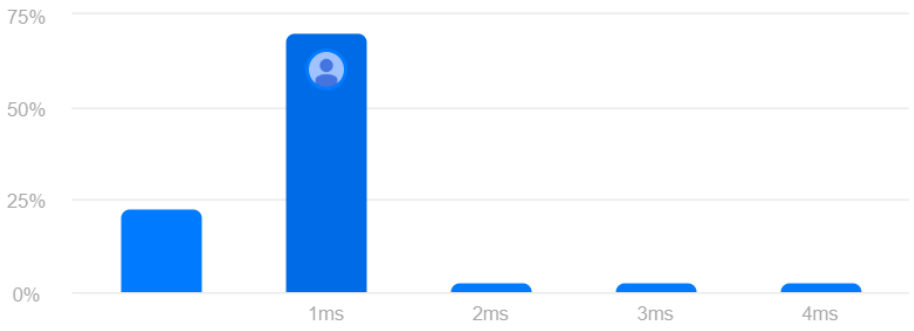
Analysis Solution

Runtime

1 ms | Beats 77.40%

Memory

46.88 MB | Beats 24.88%



Time Complexity : O(n)

Space Complexity : O(1)